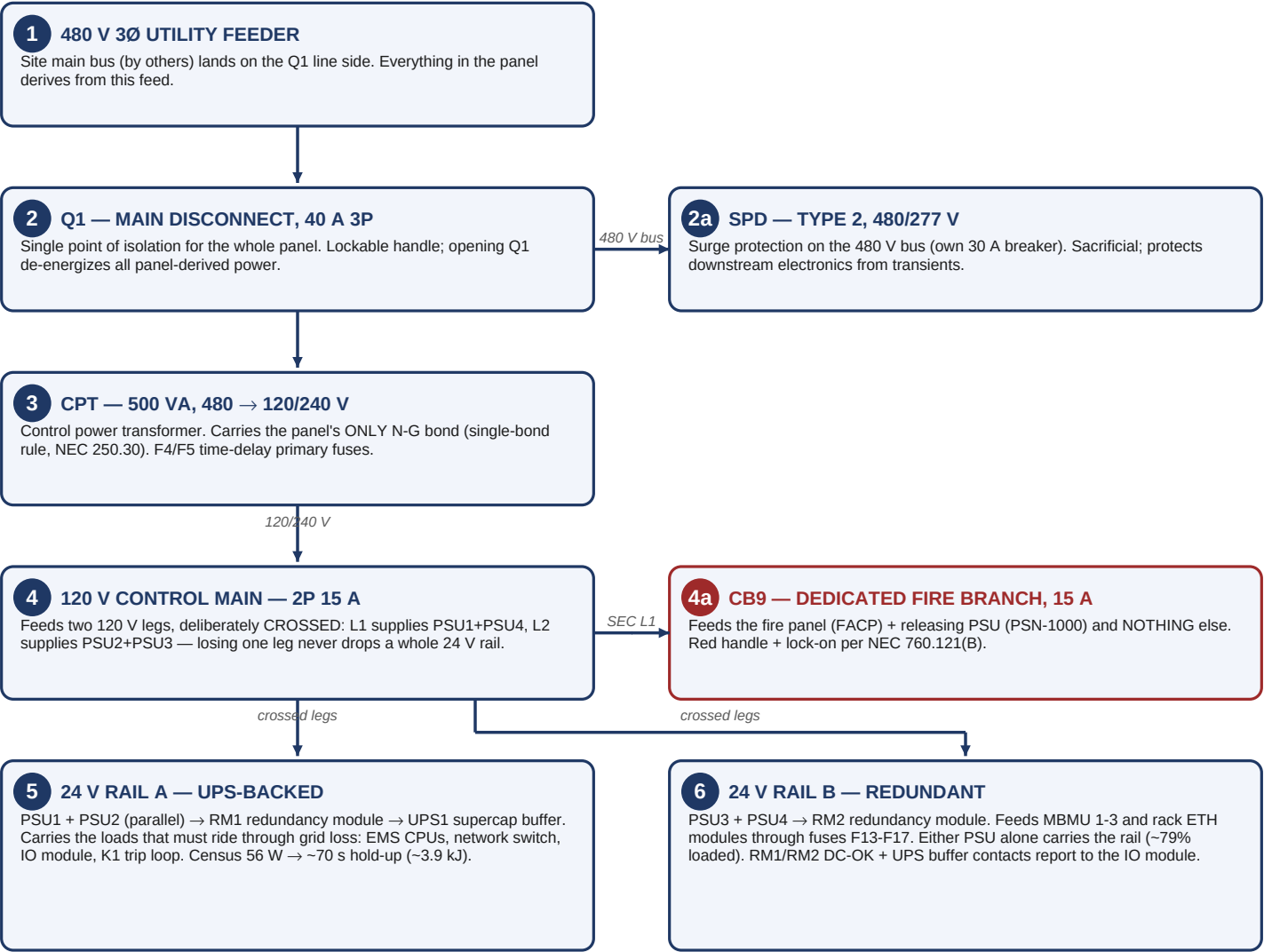
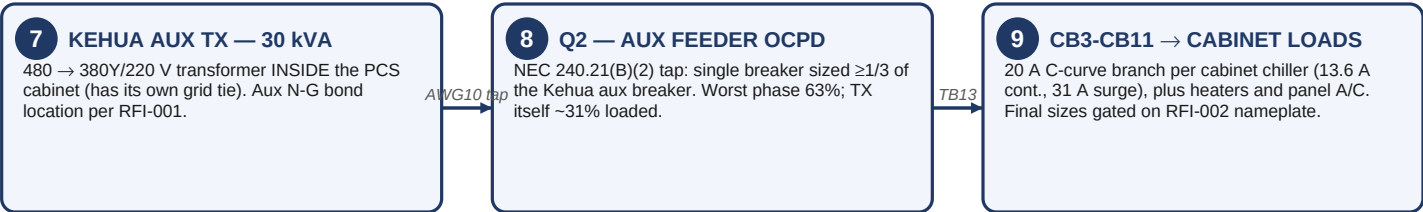


STEP BY STEP — CONTROL POWER



AUX POWER TO CABINETS (380Y/220 V)



DETECTION

1 CABINET DETECTION
Heat + smoke detectors in each EnerOne+ cabinet; supervised circuits. Cross-zone logic in the FACP prevents a single-detector release.

2 SLC — CLASS A LOOP
Panel → cab1 → cab2 → cab3 → back to panel (send and return in separate raceways). Monitor modules MM1-MM3 put the abort, manual-release and DS1 status on the same supervised loop.

3 FACP — POTTER ARC-100
Releasing-listed fire panel. Powered from CB9 + 2× 12 V 8 Ah batteries (NFPA 72 worksheet: 4.73 Ah required). Field stations: X24 abort (hold-to-abort) and X25 manual release. Abort holds the RELEASE only — it never holds the PCS trip.

5 STATUS TO EMS (TELEMETRY ONLY)
FACP alarm / trouble / supervisory contacts land on IO module inputs DI0-DI2 (N/C — open = event). The EMS sees the event but is NOT in the trip or release path.

THREE OUTPUTS ON ALARM

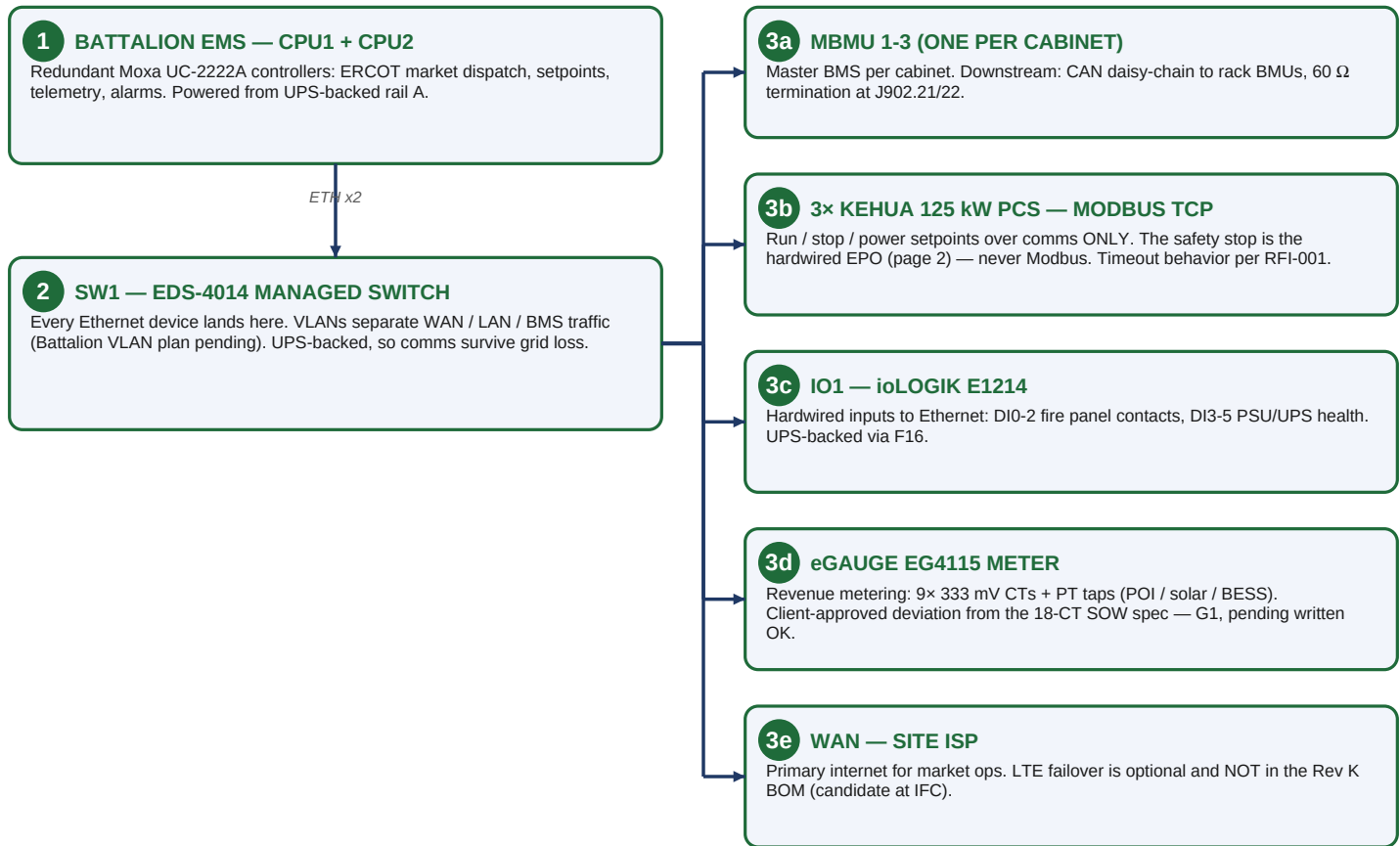
4a PCS TRIP (FAIL-SAFE)
FACP relay 26/27 is HELD ENERGIZED when healthy → F18 fuse (24 V UPS rail) → K1 relay → X23 Kehua EPO input. Alarm OR any power/wire loss drops K1 → PCS stops. Hardwired; both contact senses provided pending RFI-001.

4b AEROSOL RELEASE
FACP TRIG1-3 → PSN-1000 releasing power supply (own 2× 12 V batteries) → REL1-3 circuits → DS1 lockable, supervised service disconnect → TB14 → JX1 pins 3/4 → cabinet aerosol initiator fires. EOL resistors supervise every leg; pins confirmed via RFI-002.

4c NOTIFICATION
FACP NAC → siren + strobe (CA-6075) with EOL — audible + visual alarm at the cabinet line-up. Secondary-power duration (2 h question) flagged to the AHJ.

6 SERVICE / MAINTENANCE
Locking DS1 open disables the RELEASE for cabinet work — supervised, so the FACP shows a supervisory condition the whole time. The PCS trip chain stays armed.

STEP BY STEP — DATA PATHS



OPEN ITEMS YOUR REVIEW SHOULD CONFIRM

- **RFI-001 (Kehua)** — EPO terminal + contact sense, aux breaker ratings, N-G bond location, Modbus timeout behavior.
- **RFI-002 (CATL/Potter)** — JX1 aerosol pin functions + initiator ratings, chiller RLA/MCA/MOCP, DS1 listed device.
- **SCCR** — Utility fault current + UL 508A SB4 worksheet before fabrication.
- **NFPA 72** — Secondary-power duration — does the 2 h term apply? 8 Ah installed fails a 2 h term (needs AHJ ruling).
- **G1 — metering** — SOW says EG4130 + 18 CTs; design carries one EG4115 + 9 CTs. Needs written client approval or a redraw.
- **G2 — FACP cooling** — Recommend adding heat-pipe + 24 V fan spot cooling for the fire panel (enclosure heat balance).
- **PE stamp** — This chart and the AI review boards are working aids only — nothing substitutes for the PE of record.